Soft X-ray Spectral Diagnostics of M-Class Solar Flares Using SoLEXS/Aditya-L1

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Abstract

We present a statistical analysis of 47 M-class (M5-M9.9) solar flares observed by the Solar Low Energy X-ray Spectrometer (SoLEXS) onboard Aditya-L1. We performed spectral fitting using the CHIANTI-based isothermal model to derive flare plasma parameters (temperature and emission measure). Peak plasma temperature (T_{peak}) ranges from ~16 to 25 MK (mean 20.0 ± 2.1 MK) while peak emission measure (EM_{peak}) spans ~(1150–2800) × 10^{46} cm^{-3} (mean 1694 ± 415 × 10^{46} cm^{-3}). The EM_{peak} correlates strongly (ρ_s = 0.80) with GOES peak flux, following $EM_s = 10^{53.34} \times F_G^{(0.98 \pm 0.12)}$, while T_{peak} shows no significant correlation with GOES class or emission measure within this narrow M-class sample. In 46 of 47 flares, we observe a consistent temporal sequence: $t(T_{peak}) < t(SXR_{peak}) < t(EM_{peak})$, with median lags of $t(SXR_{peak}) - t(T_{peak})$ = 4.0 min and $t(EM_{peak}) - t(SXR_{peak})$ = 1.3 min, consistent with the chromospheric evaporation scenario. Using a 1:1 lag criterion, we classify 39 flares (~83%) as EM-controlled and 7 (~15%) as T-controlled, indicating that SXR emission is predominantly governed by emission measure rather than temperature. The $t(EM_{peak}) - t(T_{peak})$ lag correlates positively with flare duration (ρ_s = 0.82) and negatively with peak temperature (ρ_s = −0.45). These results demonstrate the capability of SoLEXS to reliably constrain flare plasma diagnostics.

Introduction

- A sudden, intense burst of radiation from the release of magnetic energy in the Sun's atmosphere, usually near active regions where field lines are twisted.
- Phases: pre-flare, impulsive and decay
- Reconnection accelerates particles that hit the chromosphere (hard X-rays); heated plasma evaporates into post-flare loops (soft X-ray and EUV).





Standard Flare Model
(A L Lysenko et al 2020)

Motivation & Objectives

- Most flare plasma studies rely on GOES/XRS: two broadband channels, no spectral information. Its flux-ratio method often underestimates temperature and emission measure.
- RHESSI spectroscopy reported flare temperatures higher than GOES, but its low-energy resolution is limited.

SoLEXS on Aditya-L1 (2–22 keV) provides high-resolution soft X-ray spectra for reliable thermal diagnostics of flares

1. Derive **T** and **EM** from SoLEXS spectra.
2. Study the temporal evolution of T, EM and SXR flux.
3. Compare the estimates and their scaling with GOES class against earlier GOES/RHESSI results.
4. Understand the role of chromospheric evaporation in flare SXR emission.

Instrument & Dataset

- The **Solar Low Energy X-ray Spectrometer (SoLEXS)** is a soft X-ray spectrometer aboard Aditya-L1, to measure the solar soft X-ray flux to study flares.
- **Sun as a Star Spectrometer** (intergrated flux)

Energy Range	2 - 22 KeV
Resolution	170 eV @5.9 keV
Cadence	1 sec
Detectors	2 SDDs
Channels	340

- **Data products:** Good-Time-Interval (.gti), 1 s light curves(.lc) and calibrated 1 s spectra (.pi) for each day.
- **Dataset: 47 flares**, GOES class **M5–M9.9** (July 2024 –May 2026). Events were selected from the GOES flare catalogue having single X-ray peak; each was cross-checked with the SoLEXS lightcurve.

Methodology

Data Preparation
Event Catalogue → Acquire SoLEXS Data → Identify Flare in SoLEXS light curve

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Spectra Generation
60s spectra → Binning with XSPEC grppha

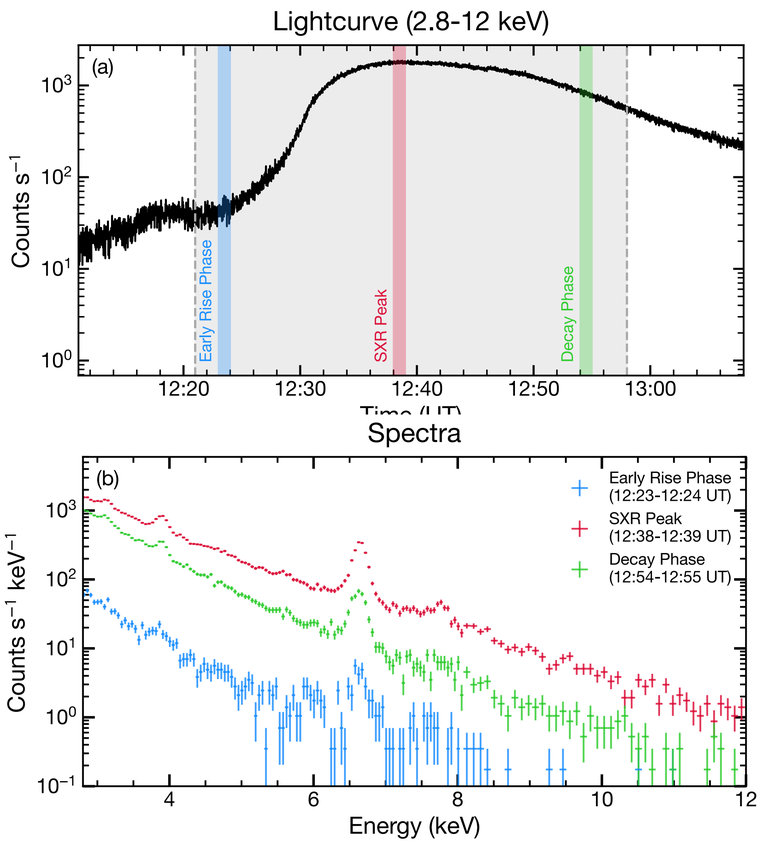
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Spectral Fitting: XSPEC + 1T chisoth Model
2.8 keV to an upper limit set by SNR, or fixed at 12 keV to limit non-thermal contribution

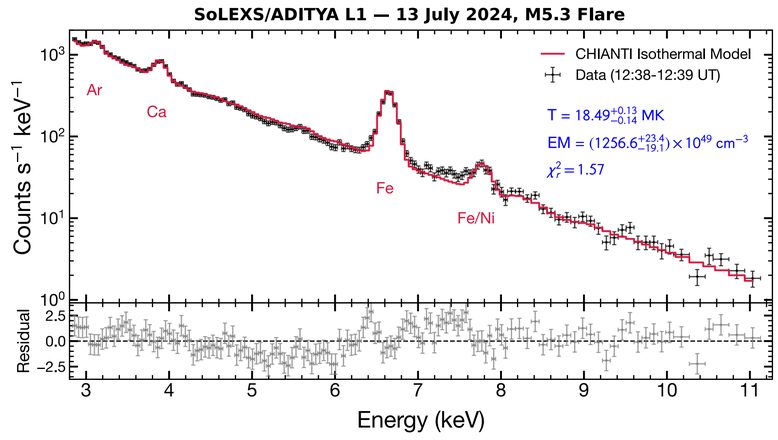
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Outputs per spectrum
T, EM and **Ar, Ca, Fe, Ni abundances** (free only when their lines are strong and within the fitting range set by SNR; otherwise fixed to coronal values)

Spectral Fitting



M5.3 Flare, 13 July 2024. (a) 2-12.8 keV lightcurve; (b) spectra corresponding to the early rise phase (blue), SXR peak (red), and decay phase (green)

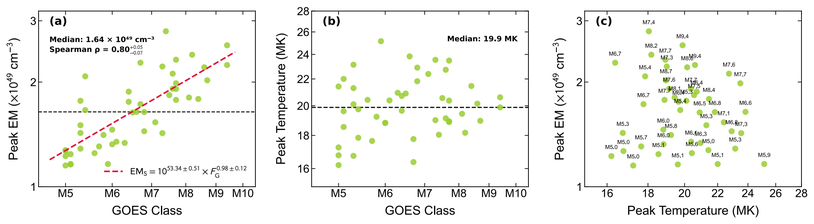


SoLEXS spectra (black) fitted with a CHIANTI isothermal model (red), showing prominent emission features from **Ar (~3.1 keV), Ca (~3.9 keV), Fe (~6.7 keV), and the Fe/Ni complex (~8 keV)**.

Peak Plasma Parameters

47	19.9 MK	1.64 x 10^{49} cm^{-3}
M5-M9.9 Flare	Median T_{peak}	Median EM_{peak}

Parameter	Min	Max	Mean	Median ± MAD
T_{peak} (MK)	16.2	25.2	20.0	19.9 ± 1.3
EM_{peak} (10^{49} cm^{-3})	1.15	2.80	1.69	1.64 ± 0.32
Duration (min)	10.0	76.0	30.1	27.0 ± 10.0
$t_{SXR} - t_T$ (min)	0.5	24.4	5.3	4.0 ± 2.0
$t_{EM} - t_{SXR}$ (min)	−0.8	13.6	2.4	1.3 ± 1.0
$t_{EM} - t_T$ (min)	1.0	38.0	7.8	6.0 ± 3.0



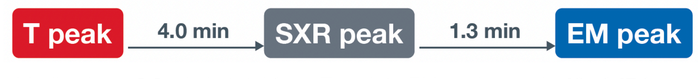
(a) Peak EM and (b) peak T against GOES class; (c) peak EM against peak T

- EM scales with flare class: $(\rho = 0.8)$, showing a relation:

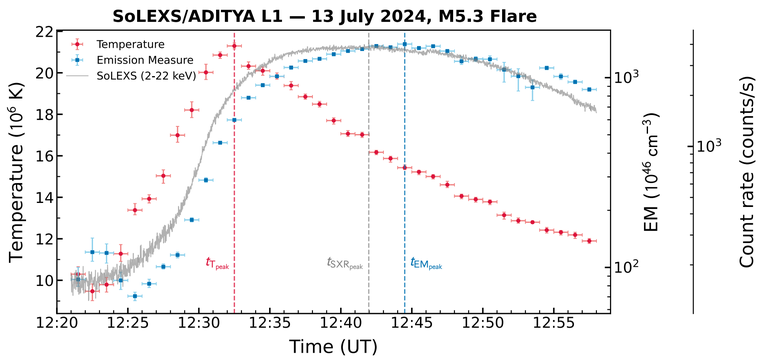
$$\mathrm{EM_S} = 10^{53\pm0.51}\, F_G^{0.98\pm0.12}\, cm^{-3}$$

- T_{peak} does not depend on class within M5–M9.9, and *T* and EM are uncorrelated.

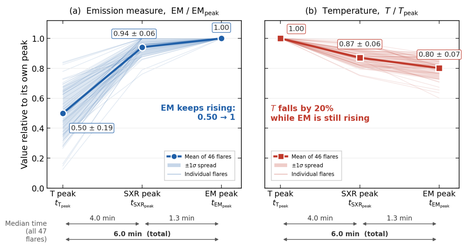
Temporal Evolution



Median lags; 6.0 min in total from *T* peak to EM peak seen in **46 of 47** flares

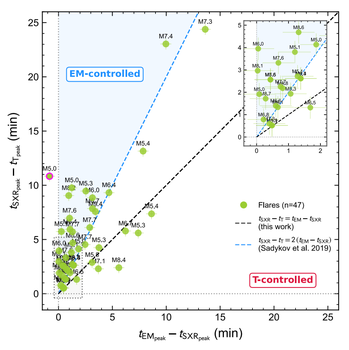


Evolution of **T**, **EM** and SoLEXS 2–22 keV count rate (grey) for the M5.3 flare.



EM/EM_{peak} and T/T_{peak} at the three peak times for all flares. At the T_{peak}, EM is only ~50% of its maximum; by the EM_{peak}, T has dropped by only ~20%.

EM -vs T- Controlled Flares

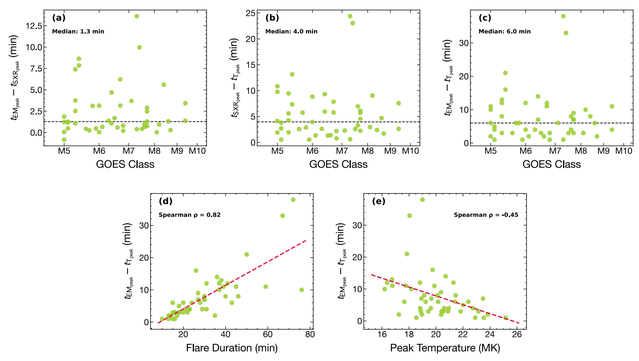


Classification Criteria: $t_{SXR_{peak}} - t_{T_{peak}} = t_{EM_{peak}} - t_{SXR_{peak}}$

39 EM controlled Flares
$t_{EM_{peak}} - t_{SXR_{peak}} < t_{SXR_{peak}} - t_{T_{peak}}$

7 T controlled Flares
$t_{SXR_{peak}} - t_{T_{peak}} < t_{EM_{peak}} - t_{SXR_{peak}}$

The SXR Peak is set mainly by the EM evolution



Lags show no dependence on GOES class. (d) The *T*–EM delay grows with flare duration (ρ = 0.82). (e) The delay against peak *T*.

Summary

- 46 of 47 flares follow $t_{T_{peak}} < t_{SXR_{peak}} < t_{EM_{peak}}$ reflecting chromospheric evaporation process.
- Most flares (39) are EM-controlled. SXR flux being predominant by EM rather than tempertaure evolution.
- Peak Emission shows strong correlation with GOES flare class $(\rho = 0.8)$ with a scaling relation:

$$\mathrm{EM_S} = 10^{53\pm0.51}\, F_G^{0.98\pm0.12}\, cm^{-3}$$

- $t_{EM_{peak}} - t_{T_{peak}}$ correlates strongly $(\rho = 0.82)$ with flare duration. Showing the loop filling takes longer for longer flares which is due to the longer flaring loops.

Future Prospects

- Wider sample (A–X class): where do flares switch between T- and EM-controlled?
- Loop lengths (AIA/STIX): does the T–EM delay scale with loop length?
- Stellar flares: the delay could estimate loop sizes in unresolved stellar flares.
- Track elemental abundance during flare evolution.

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